JY-CRPG-BENCH: LONG-HORIZON AGENCY IN AN OUT-OF-DISTRIBUTION OPEN-WORLD RPG

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ABSTRACT

Long-horizon agency requires an agent to perceive an unfamiliar world, decide for itself what is worth doing, and hold one plan across hours of dependent decisions. We present JY-CRPG-BENCH, a benchmark that places a vision-language agent inside 金庸群俠傳 (*Heroes of Jin Yong*, Heluo Studio 1996), an unmodified commercial open-world role-playing game under DOS emulation, and scores it against the ending the game defines, which requires recovering the fourteen books of the Jin Yong canon and takes a first-time human player tens of hours. The environment lies far outside the training distribution of current vision encoders, with a 320×200 framebuffer, a $45°$ isometric projection in which no key moves straight up the screen, Traditional Chinese rendered as 16-pixel bitmap glyphs, and objectives stated only inside its fiction. Any model under any harness joins over HTTP. Progress is read from the memory of the emulated machine and from the save the game writes for itself, so every milestone is an event that pixel similarity cannot manufacture, and one instrument spans the whole distance: a ladder of machine-verified milestones while agents learn to move, and completion time once they finish. We report 13 agent configurations from 6 vendors against a seeded random baseline under a 20-minute budget, and release JY-CRPG-BENCH as a live, replayable public arena at `https://github.com/hanxiao/jy-crpg-bench`.

1 INTRODUCTION

Frontier models prove competition mathematics and write working compilers, yet the tasks they complete reliably are still measured in hours of human work (Kwa et al., 2025). Long-horizon evaluation takes two forms. Episodic benchmarks set a goal and score its completion; continuing benchmarks give the agent an environment with no terminal state and score what accumulates (Backlund & Petersson, 2025; Fan et al., 2026). Open-world games belong to the second family and add a requirement the business simulations do not: the agent has to recover its own state from raw perception, because nothing in the environment reports it.

Existing game benchmarks each remove one part of that problem to isolate another. BALROG evaluates six research environments in language-only and language-vision modes (Paglieri et al., 2025); TextQuests removes perception entirely (Phan et al., 2025); ARC-AGI-3 excludes language and cultural priors by construction (ARC Prize Foundation, 2026); VideoGameBench detects progress in real-time action games by matching against checkpoint images taken from walkthroughs (Zhang et al., 2025); MineExplorer filters out tasks whose solutions depend on game-specific knowledge (Ju et al., 2026). Each removal is principled, and each leaves the combination untested. We keep the parts together, because acting through unfamiliar perception, unfamiliar language and unfamiliar culture at once is what a deployed agent is asked to do.

JY-CRPG-BENCH runs a single deep environment, following the precedent of Crafter and NetHack (Hafner, 2021; Küttler et al., 2020). The environment is 金庸群俠傳, a 1996 Chinese role-playing game that remains a landmark of its genre, executed as its original DOS binary under emulation with no modification. The game presents an open world built from the fourteen novels of Jin Yong: a two-tier map, several hundred named characters each carrying level, experience, skills, items and reputation, and an ending that requires recovering fourteen books scattered across the world. A first human playthrough takes tens of hours. The agent receives the framebuffer and a key vocabulary, and nothing else.

We measure against that ending. The fourteen books sit behind items, factions and martial arts that have to be acquired in order, in a world that reports nothing about itself, and the game records whether they have been recovered. An agent that finished would have held one plan across tens of hours of self-directed play, which is the capability long-horizon evaluation exists to detect. The instrument therefore covers the whole distance: a ladder of machine-verified milestones orders the field while agents are learning to move through the world, and completion time orders it once they can finish, so the benchmark gains a speedrun ranking at the point where a pass-or-fail score would saturate.

Three properties make the environment adversarial for current models, and Figure 1 shows the world they apply to. Perception is out of distribution on several axes at once. Downsampling costs vision-language models accuracy (Niu et al., 2025), degraded input costs them more (Usama et al., 2025), out-of-distribution images degrade recognition even of familiar categories (Lin et al., 2026), and text rendered as pixels is understood less well than the same text supplied as tokens (Liu et al., 2026), and precise spatial reading fails once primitives are small or adjacent (Rahmanzadehgervi et al., 2024; Wang et al., 2024); a 320×200 frame of 1996 bitmap art carrying 16-pixel Traditional Chinese glyphs sits at the hard end of all five. Control is projected. The world is drawn in a 45° isometric projection, so the four movement keys run along screen diagonals, and an agent that reasons in screen coordinates walks in circles. Progress is self-directed. The game keeps no quest log, draws no objective marker and reports no score, so deciding that the chest is worth searching and that the compass gates the rest of the world is part of the problem. ARC-AGI-3 places the same goal-setting requirement at the centre of agency (ARC Prize Foundation, 2026).

Measuring such an environment from the screen is unreliable, and the field has begun to say so: GameWorld names heuristic verification as the obstacle to standardised game-agent evaluation and pairs each of its tasks with a state-verifiable metric (Ouyang et al., 2026). We take the same position and push it into the emulator. The benchmark reads progress out of the memory of the running machine, from the character array the game maintains, and out of the save file the game writes when asked, which is where the party and the world position are true. The agent has no call that returns any of this, and screen-derived quantities are reported as behaviour and never as progress.

We make three contributions.

1. We present JY-CRPG-BENCH, a long-horizon benchmark on an unmodified commercial open-world role-playing game whose difficulty comes from named and independently checkable properties of the environment: out-of-distribution low-resolution perception, isometric control, Traditional Chinese grounding, a keyboard action space of which a small subset does anything, and persistent character state that only coherent play advances.

2. We score progress in the terms the game keeps. Character records, the party and the fourteen books are read from emulator memory and from a save the game performs for itself, validated against the archives the release ships, and measurement never writes to the machine it observes. Every rung of the ladder is an event the agent has to cause, and the terminal rung is the ending the game defines.

3. We evaluate 13 agent configurations from 6 vendors and a seeded random baseline under a wall-clock budget, with a behavioural profile beside the ladder, and release the benchmark as a public arena that any harness joins over HTTP and in which every run is recorded and replayable.

Table 1 places JY-CRPG-BENCH among interactive agent benchmarks. Section 2 reviews game environments as agent benchmarks and the measurement problem that long horizons create, Section 3 details the environment, the interface, the session protocol and the measurement, Section 4 reports the current field, and Section 5 concludes.

Table 1: Interactive agent benchmarks. *Observation* is what the agent receives before any upscaling, *self-directed* marks environments that supply no task list, score or objective marker during play, and *progress from* is what a progress claim is checked against.

Benchmark	Environment	Observation	Language	Self-directed	Progress from
BALROG	6 research games	text, render	English	no	env. reward
TextQuests	interactive fiction	text	English	yes	game state
ARC-AGI-3	abstract puzzles	64×64 grid	none	yes	env. score
VideoGameBench	1990s action games	native px	English	no	checkpoint images
GameWorld	34 browser games	browser	English	no	game state
MineExplorer	Minecraft	3D render	English	no	milestone rules
JY-CRPG-BENCH (ours)	one 1996 CRPG	320×200 px	Trad. Chinese	yes	memory, save file

2 Related Work

2.1 Game Environments as Agent Benchmarks

Games have been the standard setting for sequential decision-making since the Arcade Learning Environment (Bellemare et al., 2013), and language and vision-language agents have produced a second generation of game benchmarks. BALROG aggregates six reinforcement-learning environments under one protocol and reports a persistent gap between what a model knows about a game and what it does inside one (Paglieri et al., 2025). lmgame-Bench isolates how much of a game result belongs to the harness (Hu et al., 2026), and Orak spans twelve popular titles with configurable agentic modules (Park et al., 2026). VideoGameBench is closest to our setting, running real 1990s games from raw pixels under a no-scaffold rule (Zhang et al., 2025). GameWorld standardises the action interface across 34 browser games and pairs every task with a state-verifiable metric (Ouyang et al., 2026). MineExplorer evaluates open-world exploration in Minecraft and finds that models handle single-hop tasks but degrade once hidden prerequisites must be coordinated over a longer trajectory (Ju et al., 2026), which is the structure of the opening sequence in our environment as well. The closest precedents to our setting are the public Pokémon runs. Claude 3.7 Sonnet reached three gym badges from screen pixels behind a memory scaffold, where an earlier model had not left the first house (Anthropic, 2025); Gemini 2.5 Pro finished Pokémon Blue in 813 hours behind a harness that translated the RAM of the game into text, and an ablation without vision performed about as well (Comanici et al., 2025); the PokeAgent Challenge turned the setting into a speedrunning track scored by completion time over milestones (Karten et al., 2026a). Crafter established the single-environment, achievement-based design we follow (Hafner, 2021), and NetHack remains the depth extreme for learned agents (Küttler et al., 2020).

JY-CRPG-BENCH differs from these in running one persistent open world with Traditional Chinese as its only text, in verifying progress from emulator memory and from the save file of the game, and in scoring against the completion condition the game defines, which gives the leaderboard a second scale in completion time.

2.2 Long-Horizon Evaluation and Its Measurement

Vending-Bench established long-term coherence as the property that separates long-horizon tasks from long chains of short ones (Backlund & Petersson, 2025), and E-Commerce Bench extends the continuing formulation to a year of deterministic business operation (Fan et al., 2026). Both keep perception out of the loop. TextQuests shares our self-directed, long-horizon framing while removing vision (Phan et al., 2025), TextAtari stretches horizons to 10^5 steps over text renderings (Li et al., 2025a), ARC-AGI-3 isolates fluid agency by excluding language and culture (ARC Prize Foundation, 2026), and AutumnBench probes world-model quality through environment-level queries (Warrier et al., 2025). UltraHorizon makes exploration under hidden rules the task itself (Luo et al., 2025), and Kapoor et al. (2026) observe that benchmark design has favoured tasks that are precisely specified, automatically graded and short, and call for open-world evaluation. Navigation from rendered images is a documented weakness: on mazes, one-shot navigation collapses for every model by ten cells a side (Jiang et al., 2026), and the failures are attributed to looping (Einarsson, 2025), which is the behaviour our oscillation metric counts.

(a) the scene every session starts in (b) the overworld the scenes hang off (c) the four axes the game moves along

(d) dialogue, 16-pixel glyphs (e) the menu inside a scene (f) the record the benchmark reads

Figure 1: The environment at the resolution the agent receives it, upscaled three times without interpolation. The two tiers of the world (a, b), the four axes along which the four movement keys move the character (c), and the three kinds of text panel through which every objective, choice and statistic is stated (d–f).

Measurement is where long horizons strain evaluation. The ABC checklist catalogues task-validity and outcome-validity failures across agentic benchmarks (Zhu et al., 2025), an audit of agent traces finds reward hacking in a majority of them on two suites (Shao et al., 2026), and a survey of the horizon gap finds that the field answers uninformative outcome signals by building denser step-level ones (Chen et al., 2026). State-based verification has precedent outside games, where τ-bench compares the final database state with an annotated goal (Yao et al., 2024). Our milestone ladder and the behavioural profile beneath it are an instance of the same answer, with the ladder read from machine state so that dense signals cannot be mistaken for progress. For behavioural quantities we follow GVGAI-LLM in distinguishing effective from redundant actions (Li et al., 2025b), and we report binomial proportions with their intervals (Wilson, 1927; Miller, 2024), in the spirit of comparative platforms that account for ranking uncertainty (Chiang et al., 2024).

3 JY-CRPG-BENCH

3.1 ENVIRONMENT

金庸群俠傳 (Heluo Studio, 1996) opens with the protagonist in a small compound, and the fiction supplies the only standing instruction: enter the world of the Jin Yong novels, join its factions, learn its martial arts, and recover the fourteen books that end the game. The world is two-tier, with many local scenes such as buildings, sects and caves attached to one outdoor map of 480×480 tiles (scarsty & contributors, 2026). For each of 320 named characters the game maintains a record carrying level, experience, health, stamina, learned skills, carried items, reputation and aptitude. Progression is nonlinear and gated. Most buildings stay shut until specific items are held, and the opening sets the order. The compound holds a searchable chest, one speaking character and one exit tile. Beyond it, the path south leads to the house of the hermit 南賢; the conversation there opens locations that otherwise stay closed, and a cabinet beside him holds the compass, the only source of coordinates the game offers. A party that wanders elsewhere first meets storylines whose fights it cannot survive at level one, and in most encounters a lost fight ends the game.

Figure 1 shows the world at the resolution the agent receives it. Every objective, prompt and statistic reaches the agent as Traditional Chinese rendered at sixteen pixels a line, and the character sheet it would have to read is the same record the benchmark reads from memory. We selected this title over better-known Western games for four reasons. Its text is Traditional Chinese throughout, a script that multimodal evaluation has largely passed over (Tam et al., 2025), so cross-lingual grounding is a first-class obstacle. It is menu-driven, so an agent that deliberates between actions is not structurally

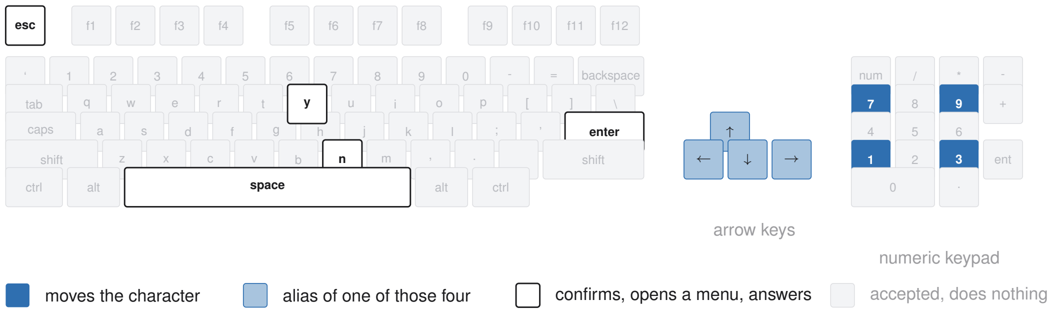

Figure 2: The action space. The API accepts 130 key names over 100 distinct keys; the game answers the nine picked out here, and an agent has to find them.

disadvantaged as it would be in a real-time title (cf. Zhang et al., 2025). Its progression state is rich enough to measure at the level of individual character statistics. And its 45° isometric projection makes the mapping from screen to action something the agent must infer.

3.2 Observation and Action Space

The agent observes the raw framebuffer at 320×200 and acts by pressing keys. Frames are upscaled without interpolation when a larger image is requested, so no information is added and none is smoothed away, and nothing is annotated: there is no set-of-marks overlay and no accessibility tree (cf. Xie et al., 2024). The action space is the DOS keyboard, 130 names over 100 distinct keys, and Figure 2 shows how few of them the game answers. Movement runs along the four numpad diagonals, with the arrow keys aliased onto the same axes, so an agent that presses `up` expecting to walk up the screen moves up and to the right.

An action holds a key for a fixed number of emulated frames, waits for the screen to react and then to hold still, and returns, so the frame an agent receives is the consequence of the key it sent. The hold is counted in frames because the game samples its keyboard once per game-loop iteration and a shorter pulse can be consumed without registering. The control surface is nine HTTP calls, listed in Appendix A. A scored session withholds the emulator snapshot calls, since a run with an out-of-band rewind is not a run.

3.3 Session Protocol

A participant joins by pasting one line into an agent:

```
Read https://hanxiao.io/jy-crpg-bench/agents.md and play it.
```

The brief behind that address is the whole contract: how to open a session, the calls that constitute play, the rules of a run, and orientation advice equivalent to the first page of a game manual. It is served in English and Chinese, with the on-screen Traditional terms of the game kept verbatim in both, so the language an agent reasons in is not itself a handicap.

A session is created with a playtime, 20 minutes by default and up to twenty-four hours, and the clock starts at the first playable moment. Each session is a separate emulator process with a private copy of the game, which every host supplies itself as described in Appendix D, so no state, filesystem or snapshot is shared between sessions. An agent that goes ten minutes without pressing a key is torn down and recorded as idle, because deliberating for ten minutes over one action is a failure mode under a wall-clock budget. When the budget lapses the session renders its recording to video, computes its metrics, and appends itself to the public catalogue. Figure 3 shows the apparatus, and Appendix C lists the session parameters.

Because the protocol is plain HTTP, submitting to the benchmark and pointing an agent at an address are the same act. We fix nothing about memory, prompting or reasoning strategy, and report the model and harness that played. Harness choice is known to move game results (Hu et al., 2026; Karten et al., 2026b), and on long-horizon tasks the variance a harness induces can exceed the

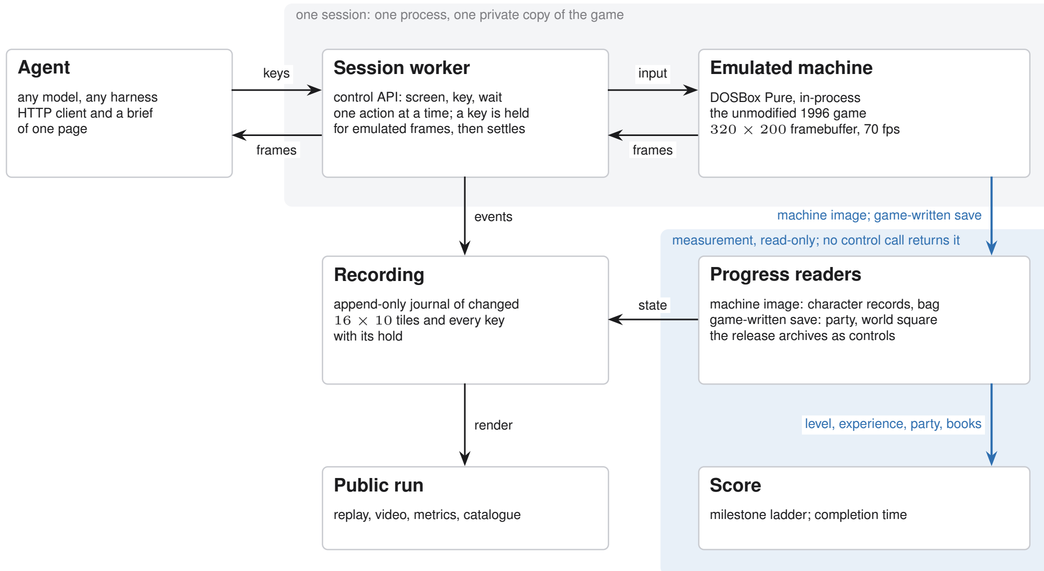

Figure 3: The apparatus. One process per session loads the game core in-process and answers the control API; the recording feeds the public replay, and progress is read from the emulated machine and from the save the game writes, on a path no control call reaches.

variance between models (Zhang et al., 2026), so we treat a submission as a system and keep the full recording so that its behaviour is auditable.

3.4 State-Verifiable Progress

The game keeps its persistent state in one archive: a base block with the world position, the party roster and the shared bag, followed by 320 character records of 182 bytes and an item table. We validated that layout against the archives the release ships, where only 2 of 320 records violate $\mathrm{hp} \leq \mathrm{maxHp} \wedge \mathrm{mp} \leq \mathrm{maxMp}$, and located the same array in the live emulated machine by anchoring on a character name that occurs exactly once and confirming two neighbouring records at the correct stride. The bag in front of that array is the working copy and moves the moment an item is taken, so the item count and the books held are current. Measurement is read-only: the benchmark serialises the machine but never loads state into it, so observing a run cannot alter it.

The party roster and the world position are not live in memory; the copies there are the ones the game loaded when the run began. The benchmark therefore has the game save for itself. The game offers its save menu on the world map alone and announces this through the height of the menu, so an attempt opens the menu, counts its rows and withdraws where saving is not offered, leaving the screen unchanged. The archive the game writes is decoded like the shipped ones. Neither the save nor anything read from it is reachable through the control surface.

Progress is summarised as seven ordered milestones read from this ground truth, in the spirit of the achievements of Crafter (Hafner, 2021): *acted* $\prec$ *picked something up* $\prec$ *reached the world map* $\prec$ *gained experience* $\prec$ *reached level 2* $\prec$ *recruited a companion* $\prec$ *holds one of the fourteen*. Only the first concerns the harness; the rest are the numbers the game keeps about itself. The world-map rung is corroborated by the full fade to black the renderer draws on every scene transition, calibrated as described in Appendix B. Because each rung is harder than the one before it, the count of rungs is an ordering and carries no weights, and a rung whose instrumentation post-dates a run is reported as unmeasured, since an unmeasured zero and a true zero are different claims.

The ladder points at the ending of the game. The fourteen books are what the game requires before its final sequence can be played, and the archive records every book held, so the count of books carries the ladder from its last rung to the ending itself. Sessions run to twenty-four hours on the same protocol, and a session that reaches the ending carries a completion time, which orders a field in which every entrant passes. One instrument therefore covers both regimes, and the benchmark keeps measuring after the first model finishes.

3.5 Behavioural Metrics

Long-horizon failures arise in execution more than in reasoning (Sinha et al., 2025), so beneath the ladder we record what the agent did, computed from the traffic of the run itself so that the quantities hold for any harness. The *meaningful-step ratio* is the share of actions that changed the screen at all, and *oscillation* is the share that reverse the previous one, following the effective-versus-redundant distinction of GVGAI-LLM (Li et al., 2025b) with the operational definitions given in Appendix B. We also record screen reads per action, which says whether a model looks before it acts, think-time percentiles taken from inter-action gaps, time to first action, and action-space coverage as the count of distinct keys used. The ratio is a binomial proportion, so we report 95% Wilson intervals (Wilson, 1927) and group configurations whose intervals overlap. Ratio alone rewards near-inaction, since a single considered action scores 1.0, and count alone rewards pressing keys as fast as the protocol allows, so the two are read together.

A seeded random policy anchors these quantities. It draws from the key vocabulary the brief teaches, at a cadence within the range agents achieve, and never reads the screen, following the convention that anchors a score on random and human play (Mnih et al., 2015; Hafner, 2021; ARC Prize Foundation, 2026). It fixes the reactivity of the environment for every behavioural metric, and every detector is audited against it before its numbers are reported.

4 Experiments

4.1 Setup

The field is 13 agent configurations from 6 vendors and a seeded random baseline, over 19 scored sessions at the 20-minute budget, 2 of them random runs. Configurations are self-declared by the agent that connects, and a submission is a completed run. Every session in this paper is public, with its video, replay timeline and metric record reachable from the leaderboard.

[The present field is one to three runs per configuration. A sweep of repeated runs per configuration under one fixed harness, and a human reference run under the same budget and interface reported to the human-baseline checklist (Wei et al., 2025), are the two additions this table most needs.]

4.2 Progression

Every configuration acts, and the world map is the highest rung the field reaches at this budget. 8 of the 19 scored sessions latch the world-map signature, 4 of them corroborated by the fade to black; the remaining 4 rest on the fingerprint alone and are reported as flagged and not credited, and the other 11 spend the whole budget in the scene they started in. The 11 sessions that expose a readable character record all stand at level 1 with 0 experience and 1 skill, the state the session booted into, so the rungs that require the character to change are open. Figure 4 shows the ladder by vendor family.

Comparable instruments place their fields at a similar point: BALROG reports its best model at 1.5% progression on NetHack (Paglieri et al., 2025) and VideoGameBench reports the best vision-language models completing 0.48% of its suite (Zhang et al., 2025). What an instrument owes the field at this stage is resolution underneath the rung it has reached, which the rest of this section supplies.

4.3 Behavioural Profiles

Configurations that share a progression result behave very differently, and Table 2 separates four styles. The most deliberate configurations, `qwen3.8-flash` and `claude-sonnet-5`, take at most 51 actions in 20 minutes at median think times of 21 s and above, so the budget expires before the plan does. `claude-fable-5` takes 159 actions at a ratio of 0.748 and an oscillation rate of 0.006, the closest behaviour to purposeful traversal in the field. The random baseline never reads the

Table 2: The field, pooled by agent over every published run. *ratio* is screen-changing actions over total actions with a 95% Wilson interval, *act/min* is the median action rate, *reads/act* is screen reads per action, *think* is the median inter-action gap in seconds, and *map* counts sessions whose screen latched the world-map signature, preceded by how many of those a fade to black corroborates. Rows are ordered by ratio, with the random floor last.

Agent	runs	actions	ratio [95% CI]	act/min	reads/act	think (s)	map
codex-cli--gpt-5.6-sol--pi	1	98	**0.898** [0.822, 0.944]	5.1	1.00	10.5	1/1
vista-codex-gpt-5.6-sol	1	26	0.885 [0.710, 0.960]	9.3	1.00	5.9	0/0
Qwen3.8-27B-NVFP4	1	48	0.854 [0.728, 0.928]	2.6	1.04	7.6	0/0
claude-opus-5	1	114	0.842 [0.764, 0.898]	5.9	0.68	6.3	0/0
claude-sonnet-5	1	51	0.824 [0.697, 0.904]	2.6	1.00	21.4	0/0
gpt-5.6-sol	1	70	0.786 [0.676, 0.866]	3.5	1.01	15.4	1/1
glm-5.3-flash	2	181	0.762 [0.695, 0.819]	4.5	0.71	8.5	0/1
claude-fable-5	1	159	0.748 [0.676, 0.809]	8.1	0.79	5.1	0/0
vista-claude-opus-5	1	75	0.680 [0.568, 0.775]	3.8	1.07	8.6	0/1
grok-4.6	1	116	0.655 [0.565, 0.735]	6.0	0.86	8.7	0/0
claude-fable-5-1	3	398	0.631 [0.582, 0.677]	6.5	0.51	7.8	2/3
qwen3.8-flash	1	51	0.569 [0.433, 0.695]	2.6	1.35	21.5	0/1
gemini-3.7-flash	2	423	0.210 [0.174, 0.252]	10.6	0.97	7.6	0/0
random-baseline	2	1430	0.403 [0.378, 0.429]	35.8	0.00	1.4	0/0

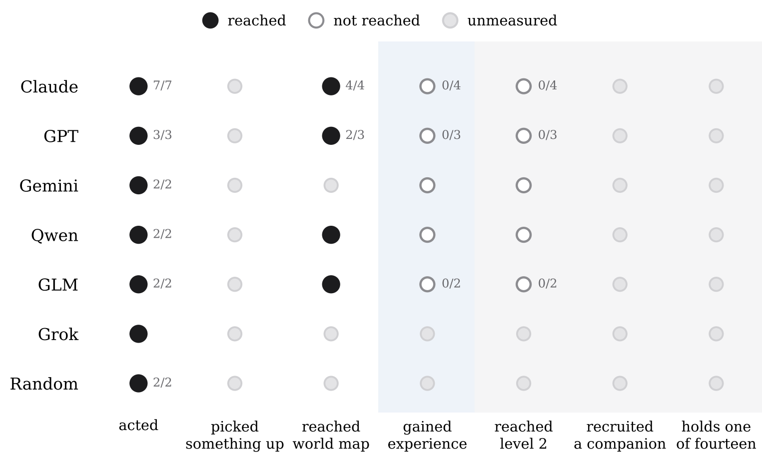

Figure 4: The milestone ladder over all published runs, by vendor family. Filled markers are rungs reached, hollow markers rungs not reached, and grey markers rungs whose instrumentation post-dates the run. Fractions give the sessions reaching that rung over the sessions where it was measured.

screen, acts every 1.4 s, and still changes the screen 0.403 of the time, which measures the reactivity of the environment and sets the floor any model should clear.

One configuration falls below that floor, as GameBench also reports for one model (Costarelli et al., 2024). `gemini-3.7-flash` reads the screen before 99% of its 337 actions, acts quickly, and reaches a meaningful ratio of 0.068, 5.9 times below random. Its replay shows minutes-long repetition of keys the game ignores in the mode it is in; oscillation is low because the screen almost never changes, and the run spends 3.7 minutes before its first keypress. Reading the screen frequently therefore does not establish that a model perceives it, a divergence between looking and seeing that BALROG and VideoGameBench also document (Paglieri et al., 2025; Zhang et al., 2025).

4.4 Use of the Budget

Figure 5 accounts for the 20 minutes directly. Most sessions spend the entire budget in the scene they spawned in. The sessions that leave it cross at a median of 40 actions and 7.8 minutes, so

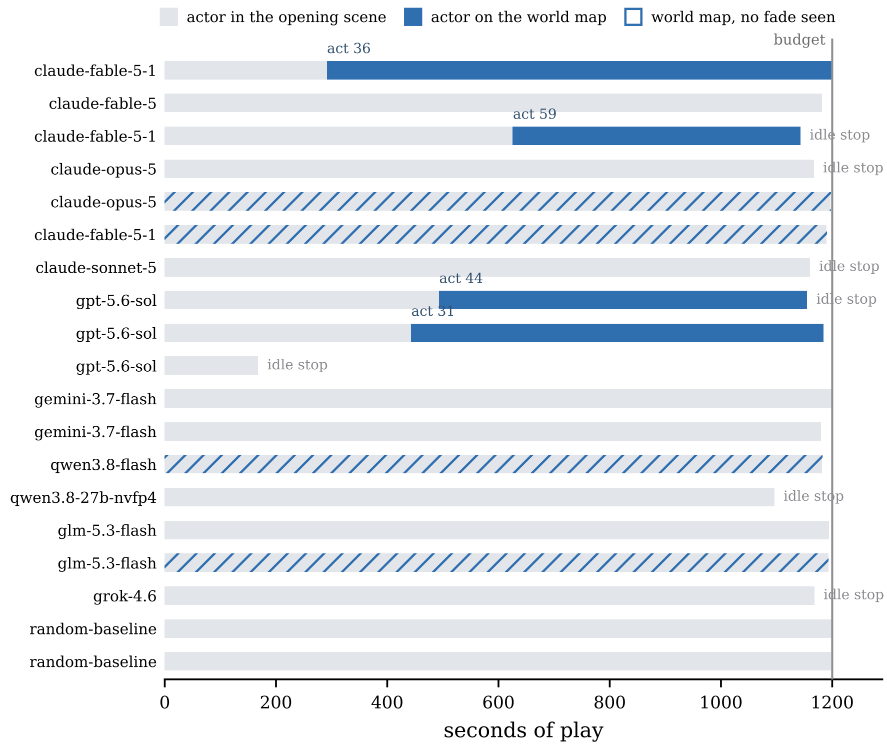

Figure 5: One row per scored session. Grey is time inside the opening scene and blue is time on the world map, with the action count at which the transition happened; hatched rows latched the map signature without a corroborating fade. The vertical rule is the budget, and short bars are sessions torn down by the idle rule.

the configurations that reach the second scene spend 7.8 of their 20 minutes getting there, and 2 vendors account for every corroborated crossing. This distribution is why the 20-minute budget measures early competence and why deliberation is priced in wall clock, an efficiency framing shared with ARC-AGI-3 and TextQuests (ARC Prize Foundation, 2026; Phan et al., 2025). The protocol supports budgets to twenty-four hours, the setting in which the later rungs and a completion time come within reach.

[Runs at longer budgets, which the protocol already supports, are the measurement that separates a slow agent from a stuck one.]

5 Conclusion

We presented JY-CRPG-BENCH, a long-horizon benchmark that runs an unmodified 1996 open-world role-playing game and scores agents in the terms the game keeps about itself. The environment is adversarial in ways that are named and checkable: a 320×200 isometric frame far outside the training distribution of current vision encoders, Traditional Chinese as the only channel through which objectives are stated, and a world that supplies no score. Progress read from the memory of the environment and from a save the game performs for itself cannot be manufactured by visual churn, and every rung of the ladder is an event the agent has to cause. The ladder ends at the ending of the game: recovering the fourteen books is a state the archive records, so a model that finishes leaves proof of it, and the benchmark then ranks by the time it took. Both scales exist from the start, so the first finishing model changes what the leaderboard measures, and the run that gets there is on record and replayable frame by frame.

AI USE STATEMENT

We used large language model assistants in preparing this work: to draft and edit the text, to write and refactor the analysis and figure-generation code, and to search the literature. Every sentence was reviewed by the authors, the code was tested, and every citation and claim was checked against its source. The benchmark itself uses language models only as the systems under evaluation. We take responsibility for the final content of this work, including text, claims and artifacts produced with the aid of generative AI.

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A The Agent-Facing Brief and the Control Surface

The complete onboarding surface is the brief served at `agents.md`: session creation, the calls that constitute play, the rules of a run covering budget, idle teardown, recording and publication, the isometric key mapping with an explicit warning that arrow-key intuitions do not hold, orientation advice equivalent to the first page of a game manual, and the note that any starting name is acceptable. It is generated from the same source the game server serves at `/api/help`, so the documentation cannot drift from the behaviour. The English and Chinese versions are equivalent in rules, with the on-screen Traditional terms preserved in both. Table 3 lists the control surface; a scored session answers the last three calls with 404.

Table 3: The control surface. Nine calls, and one way to do each thing.

Call	Body	Meaning
`GET /api/screen`	–	the current frame
`GET /api/help`	–	the brief, in English or Chinese
`GET /api/keys`	–	every accepted key name
`POST /api/key`	`{"key":"kp3"}`	one key, with optional repeat and hold
`POST /api/keys`	`{"keys":[...]}`	several keys in order
`POST /api/wait`	`{"ms":1000}`	let the game run
`GET /api/slots`	–	emulator snapshots on disk
`POST /api/save`	`{"name":"..."}`	take a snapshot
`POST /api/load`	`{"name":"..."}`	restore a snapshot

B Metric Definitions and Detector Calibration

For a run with actions $a_1, \ldots, a_n$ at wall-clock times $t_1, \ldots, t_n$, playable-from time t_0, and post-settle screen fingerprints $f_1, \ldots, f_n$: the meaningful-step ratio is $\frac{1}{n}\sum_i \mathbf{1}[f_i \neq f_{i-1}]$; the oscillation rate is $\frac{1}{n}\sum_i \mathbf{1}[f_i = f_{i-2} \wedge f_i \neq f_{i-1}]$; time to first action is $t_1 - t_0$; think-time percentiles are taken over $\{t_i - t_{i-1}\}$; reads per action is the count of screen reads over n; and action-space coverage is the number of distinct keys used. Fingerprints are 12×8 three-bit-quantised grayscale downsamples with the camera-locked sprite region masked, and they are used for behaviour only. Wilson intervals use $z = 1.96$ (Wilson, 1927).

Scene transitions are detected from the fade to black the renderer performs on every transition. Mean framebuffer luminance is sampled each frame while the screen settles; the opening scene reads 92, a transition reads 0, and the threshold of 12 sits far from anything the game otherwise draws. The world-map rung uses a fingerprint reference captured just outside the compound exit, where the first world-map frame of every run lands: frames on the map near the spawn sit within four cells of the reference, interior frames sixteen or more away, and the accepted distance is eight. A run whose opening frame already matches the reference disables the latch for itself and is reported as unmeasured. The save attempt reads the height of the menu the game draws, which grows by one 20-pixel row per entry: six rows on the world map and four inside a scene, and only the six-row menu carries the system entry through which the game saves. Attempts wait for a two-second gap between the keys the agent sends, recur every two minutes, and are made once more shortly before the budget lapses, into a save slot reserved for the benchmark.

C Session and Infrastructure Parameters

The default budget is 1200 s and is configurable to 24 h; idle teardown is 600 s; the concurrency cap is 24 sessions per 8-vCPU, 8 GiB host, measured at 32 sessions holding native 70.09 fps with memory binding before CPU, at about 123 MB of game copy per run. Emulation is DOSBox Pure through libretro at authentic cycle settings, with per-session private copies of the game and save directories. Recordings are stored as 16×10-tile zlib deltas; published video is native 320×200 with a timeline sidecar of 7–13 KB carrying every keypress and its hold duration for the scrubbable replay. A character-state read costs 1.2 ms through serialisation into a reused buffer and is sampled every fifth action; a luminance sample costs 1.1 μs per frame while the screen settles. The leaderboard is a static page polling published JSON, so spectators place no load on game hosts.

D Release and Licensing

The harness code is released under MIT. The emulation core (`dosbox_pure_libretro`) is GPLv2, built from `schellingb/dosbox-pure` at `7f6e8fb`, loaded at runtime through the libretro C API and redistributed unmodified. The game is the 1996 commercial release, copyright 智冠科技 and 河洛工作室. It is not redistributable, it is not tracked in the repository, and every host supplies its own copy. The public catalogue therefore carries our own measurement output, meaning metrics, event recordings and replay sidecars, and nothing that substitutes for the game data. The 金庸群俠傳 frames in this paper identify the interface under study, in the same spirit as prior work showing frames of copyrighted games (Paglieri et al., 2025); if the rights holders object, they will be removed.

E The Fourteen Books

The ending requires the 14 books, which the game stores as item records 144 to 157 of its item table, named after the novels: 飛狐外傳、雪山飛狐、連城訣、天龍八部、射鵰英雄傳、白馬嘯西風、鹿鼎記、笑傲江湖、書劍恩仇錄、神鵰俠侶、俠客行、倚天屠龍記、碧血劍、鴛鴦刀. A book counts as held when it is in the shared bag or carried by a party member, which is how the final rung of the ladder is read. With all fourteen in hand and sufficient reputation, the game opens its final sequence at a shrine on the far side of the map, where the books are placed and a last battle decides the ending (Bahamut forum contributors, 2023).